

COURSE MATERIAL

SUBJECT	PART-B BASIC ELECTRONICS ENGINEERING (EE23AES101)
UNIT	II
COURSE	B.TECH
DEPARTMENT	ECE
SEMESTER	1-1
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1. Course Objectives

The objectives of this course are

1. To understand the working principle of various electronic devices.
2. To acquire knowledge on the operating conditions of a DC power supply system.
3. To learn basic techniques in the design of a logic circuit.

2. Prerequisites

Students should have knowledge on

1. Applied Physics
2. Network Analysis

3. Syllabus

UNIT II

Rectifiers and power supplies: Block diagram description of a dc power supply, working of a full wave bridge rectifier, capacitor filter (no analysis), working of simple Zener voltage regulator. Amplifiers: Block diagram of Public Address system, Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response. Electronic Instrumentation: Block diagram of an electronic instrumentation system.

4. Course outcomes

After completion of this subject, students will be able to

1. **Apply the concept of science and mathematics to understand the working principles of electronic devices.**
2. **Analyze the working principle of a DC power supply system and Amplifiers.**
3. **Solve digital logic circuits and implement using different logic gates.**

5. Co-PO / PSO Mapping

EDC	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	2	1										3	2
CO2	3	3	3	2									3	3
CO3	3	3	2	2									3	3

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Block diagram description of a dc power supply	T1
2		working of a full wave bridge rectifier, capacitor filter (no analysis)	T1
3		working of simple Zener voltage regulator	T1
4		Amplifiers: Block diagram of Public Address system	T1
5		Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response.	T1
6	2	Electronic Instrumentation: Block diagram of an electronic instrumentation system.	T1
7		Elementary Treatment of Small Signal CE Amplifier.	T1

7. Activity Based Learning

1. Observing output of Rectifier outputs in the laboratory using CRO
2. Observing Physical components in Sound Amplifier System

8. Lecture Notes

2.1 Rectifiers and power supplies: Introduction

This chapter introduces the operation of power supply circuits built using filters, rectifiers, and then voltage regulators. Starting with an ac voltage, we obtain a steady dc voltage by rectifying the ac voltage, then filtering to a dc level, and, finally, regulating to obtain a desired fixed dc voltage. The regulation is usually obtained from an IC voltage regulator unit, which takes a dc voltage and provides a somewhat lower dc voltage, which remains the same even if the input dc voltage varies or the output load connected to the dc voltage changes.

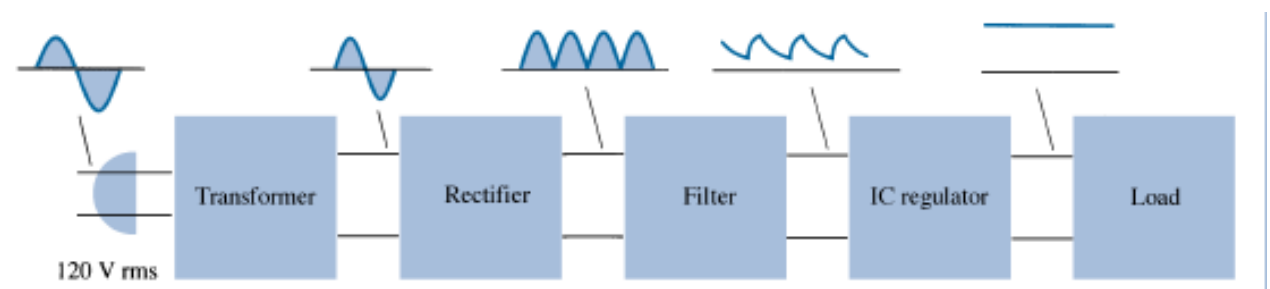
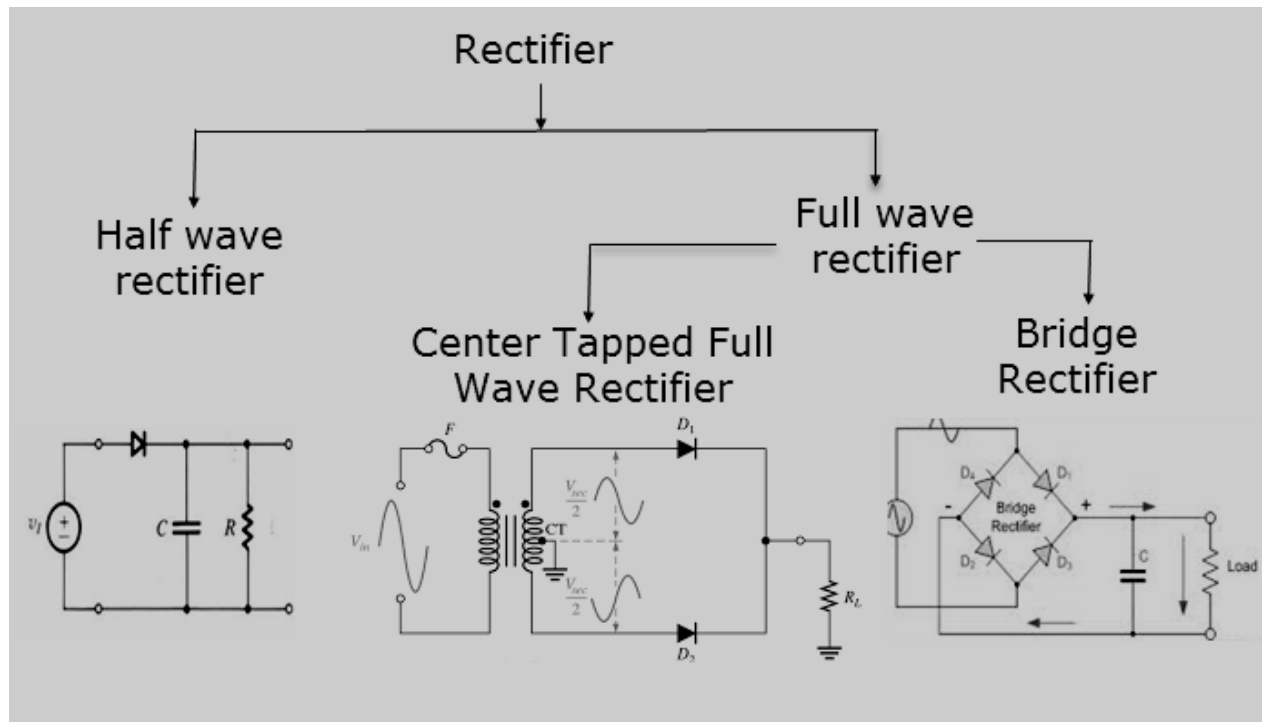


Fig 2.1 : Block diagram of a DC power supply.

A block diagram containing the parts of a typical power supply and the voltage at various points in the unit is shown in Fig. 2.1.

The ac voltage, typically 120 V rms, is connected to a transformer, which steps that ac voltage down to the level for the desired dc output.

A diode rectifier then provides a full-wave rectified voltage, which is initially filtered by a basic capacitor filter to produce a dc voltage. This resulting dc voltage usually has some ripple or ac voltage variation. A regulator circuit can use this dc input to provide a dc voltage that not only has much less ripple voltage, but also remains at the same dc value even if the input dc voltage varies somewhat or the load connected to the output dc voltage changes. This voltage regulation is usually obtained using one of a number of popular voltage regulator IC units.



2.2 Working of a full wave bridge rectifier

Rectifier:

A rectifier converts an AC signal into DC, and a bridge rectifier does this using a diode bridge. A diode bridge is a system of four or more diodes in a bridge circuit configuration, wherein two circuit branches are branched by a third. A bridge rectifier provides full-wave rectification.

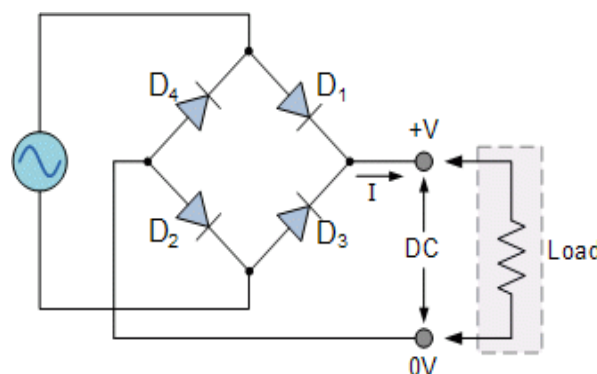
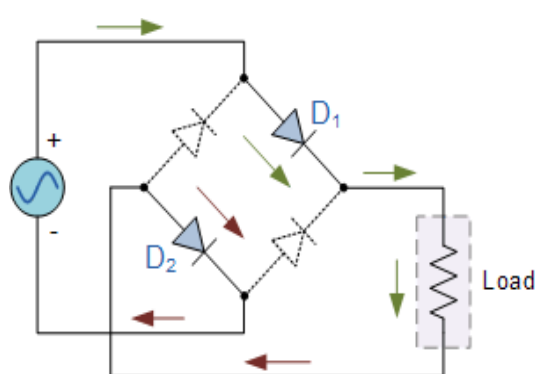


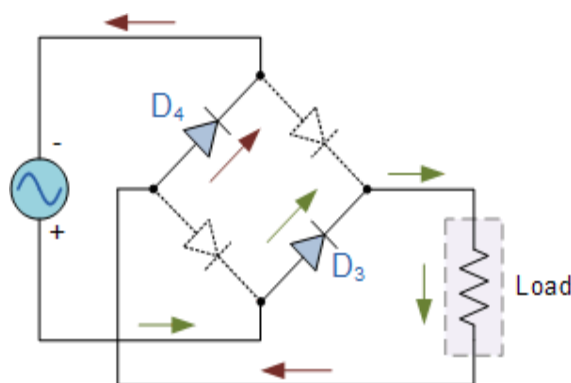
Fig 2.2 Full wave Bridge Rectifier

The four diodes labelled D1 to D4 are arranged in “series pairs” with only two diodes conducting current during each half cycle.

During the positive half cycle of the supply, diodes D1 and D2 conduct in series while diodes D3 and D4 are reverse biased and the current flows through the load. During the negative half cycle of the supply, diodes D3 and D4 conduct in series, but diodes D1 and D2 switch “OFF” as they are now reverse biased. The current flowing through the load is the same direction as before.



Positive Half-cycle



Negative Half -Cycle

As the current flowing through the load is unidirectional, so the voltage developed across the load is also unidirectional the same as for the previous two diode full-wave rectifier, therefore the average DC voltage across the load is $0.637V_{\max}$.

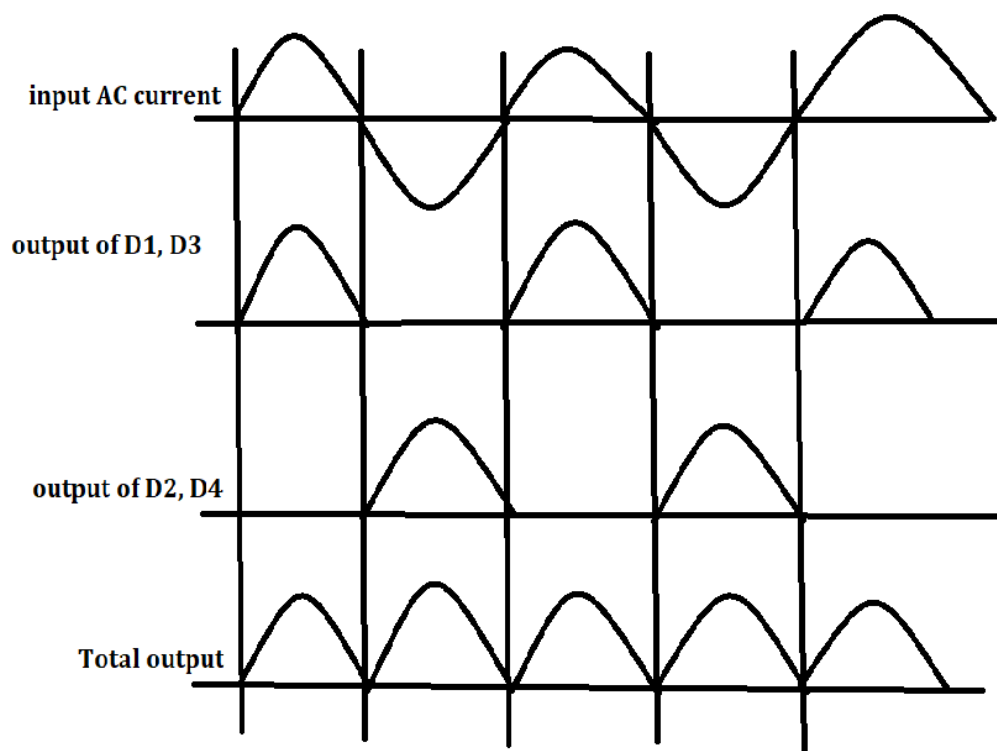


Fig 2.3: Input and output waveforms of the Full Wave Bridge Rectifier

2.3 Capacitor filter

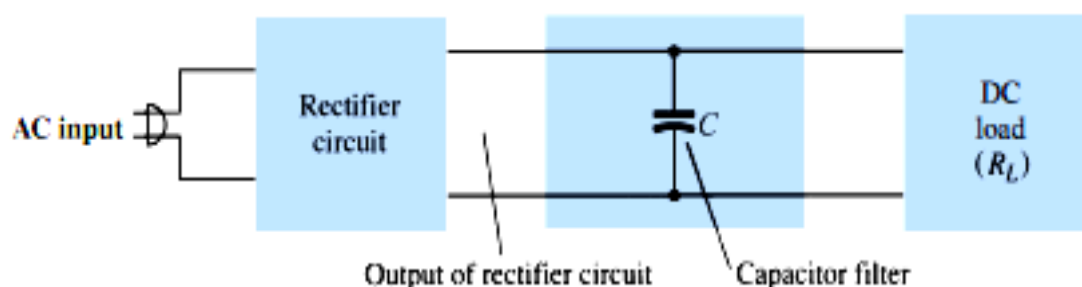
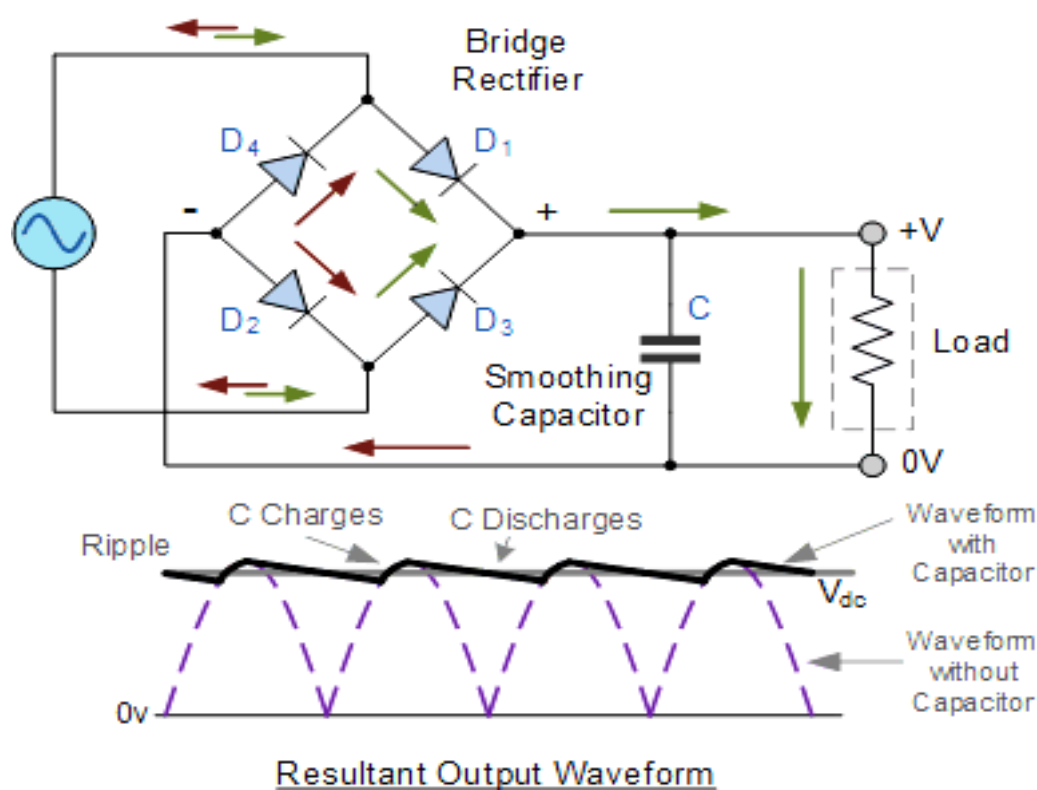


Fig. 2.4: Basic capacitor filter

The rectified output from the bridge rectifier is not pure DC. Because AC character also exists in the output. We can improve the DC output of the rectifier while at the same time reducing the AC variation of the rectified output by using smoothing capacitors to filter the output waveform. Smoothing or reservoir capacitors connected in parallel with the load across the output of the full wave bridge rectifier circuit increases the average DC output level even higher as the capacitor acts like a storage device as shown below. The smoothing capacitor converts the full-wave rippled output of the rectifier into a more smooth DC output voltage.



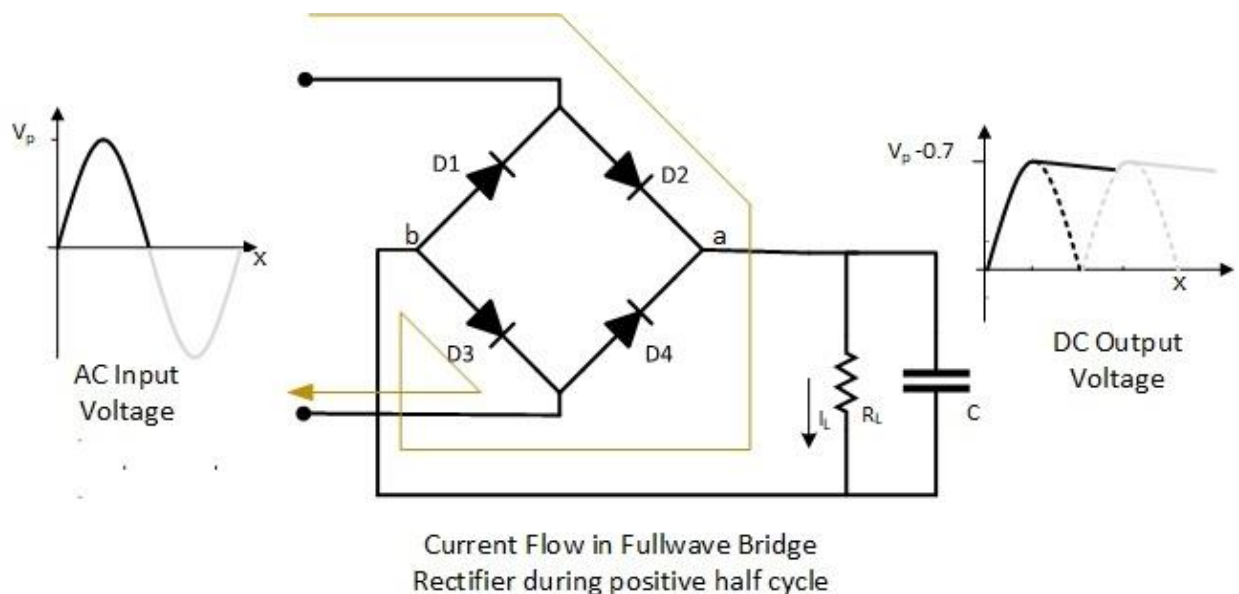
Working of Full Wave Bridge Rectifier with Capacitor Filter:

Positive Half Cycle of rectifier:

During the positive cycle of the AC input, the upper corner of the bridge is comparatively positive where diodes D1 and D2 are connected. In addition, the lower corner of the bridge is comparatively negative where diodes D3 and D4 are connected.

In this situation, the diode D2 is forward biased as its anode is connected to comparatively higher potential and diode D1 is reversed biased as its cathode is connected to the comparatively higher voltage. Similarly, at the lower corner, the diode D3 is forward biased as its cathode is connected comparatively lower voltage and diode D4 is reversed biased as its anode is connected to the comparatively higher voltage.

For the positive cycle, the current flows from the upper corner of the bridge through diode D2, then through the load resistor from point a towards point b and diode D3, completing its path to the lower corner.

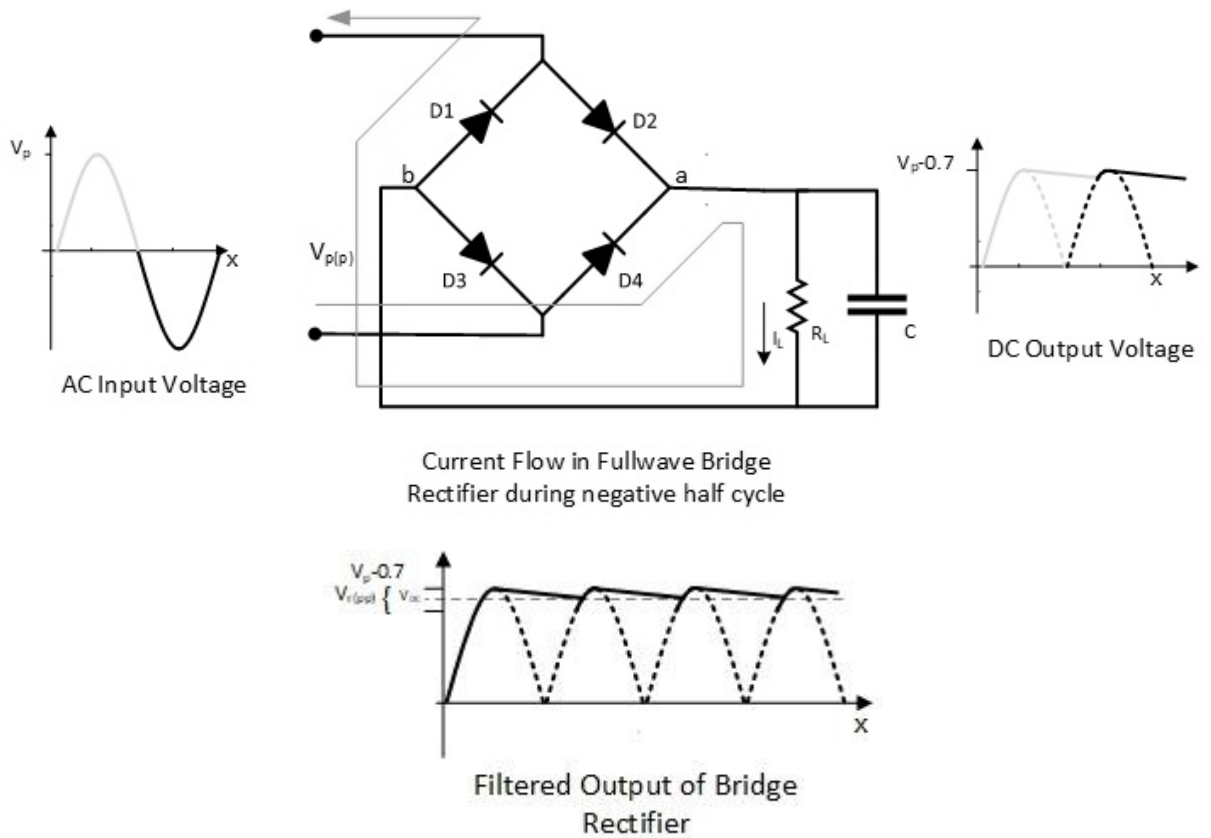


Negative Half Cycle of rectifier:

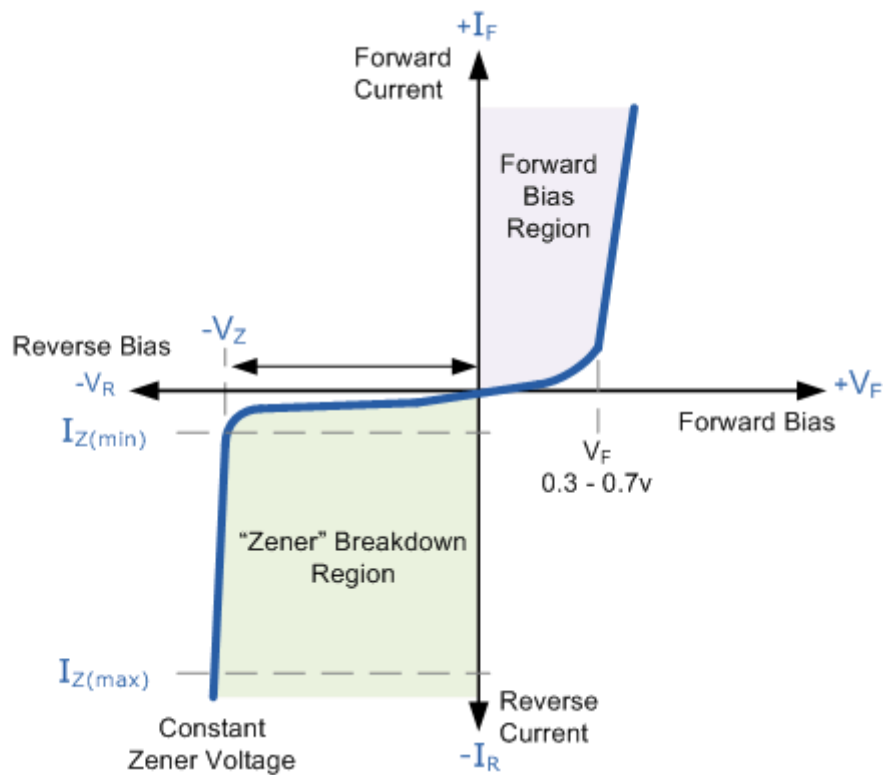
During the negative cycle of the AC input, the upper corner of the bridge is comparatively negative where diodes D1 and D2 are connected. In addition, the lower corner of the bridge is comparatively positive where diodes D3 and D4 are connected.

In this situation, the diode D1 is forward biased as its cathode is connected to comparatively lower potential and diode D2 is reversed biased as its anode is connected to the comparatively lower voltage. Similarly, at the lower corner, the diode D4 is forward biased as its anode is connected to a comparatively higher voltage and diode D3 is reversed biased as its cathode is connected to the comparatively higher voltage.

For the negative cycle, the current flows from the lower corner of the bridge through diode D4, then through the load resistor from point a towards point b and diode D1, completing its path to the higher corner.



2.4 Simple Zener voltage regulator:



The function of a voltage regulator is to provide a constant output voltage to a load connected in parallel with it in spite of the ripples in the supply voltage or variations in the load current.

The Zener Diode is used in its “reverse bias” or reverse breakdown mode. From the I-V characteristics curve above, we can see that the Zener diode has a region in its reverse bias characteristics of almost a constant negative voltage regardless of the value of the current flowing through the diode.

This voltage remains almost constant even with large changes in current providing the Zener diodes current remains between the breakdown current $I_{Z(\min)}$ and its maximum current rating $I_{Z(\max)}$.

The fact that the voltage across the diode in the breakdown region is almost constant turns out to be an important characteristic of the Zener diode as it can be used in the simplest types of voltage regulator applications.

A Zener diode will continue to regulate its voltage until the diodes holding current falls below the minimum $I_{Z(\min)}$ value in the reverse breakdown region.

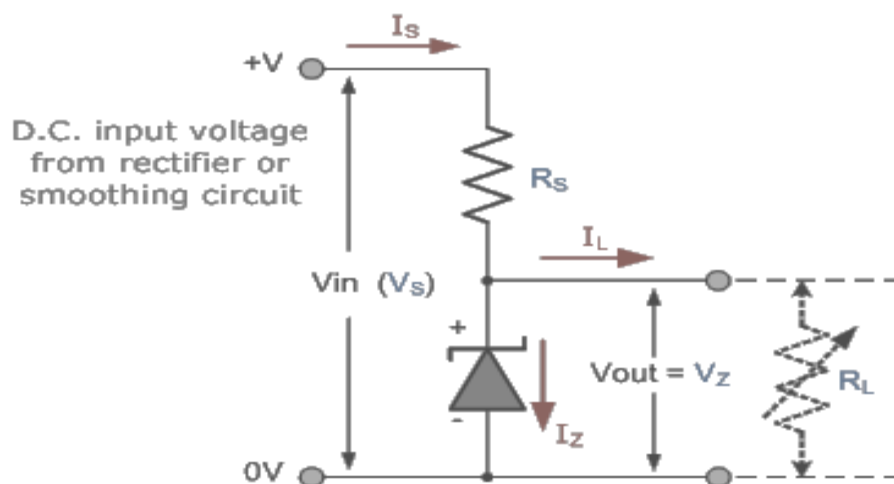


Fig 2.5: Zener Diode Regulator

The Zener voltage regulator consists of a current limiting resistor R_S connected in series with the input voltage V_S with the Zener diode connected in parallel with the load R_L in this reverse biased condition. The stabilized output voltage is always selected to be the same as the breakdown voltage V_Z of the diode.

Resistor, R_S is connected in series with the Zener diode to limit the current flow through the diode with the voltage source, V_S being connected across the combination. The stabilized output voltage V_{out} is taken from across the Zener diode.

The Zener diode is connected with its cathode terminal connected to the positive rail of the DC supply so it is reverse biased and will be operating in its breakdown condition. Resistor R_S is selected so to limit the maximum current flowing in the circuit.

With no load connected to the circuit, the load current will be zero, ($I_L = 0$), and all the circuit current passes through the Zener diode which in turn dissipates its maximum power.

Also, a small value of the series resistor R_S will result in a greater diode current when the load resistance R_L is connected and large as this will increase the power dissipation requirement of the diode so care must be taken when selecting the appropriate value of series resistance so that the Zener's maximum power rating is not exceeded under this no-load or high-impedance condition. The load is connected in parallel with the Zener diode, so the voltage across R_L is always the same as the Zener voltage, ($V_R = V_Z$).

There is a minimum Zener current for which the stabilization of the voltage is effective and the Zener current must stay above this value operating under load within its breakdown region at all times. The upper limit of current is of course dependent upon the power rating of the device. The supply voltage V_S must be greater than V_Z .

2.5 Amplifiers:

Block diagram of Public Address system

A "Public Address" system is anything that amplifies sound so more people can hear it. A simple public address system (or PA system) is shown in the following block diagram.

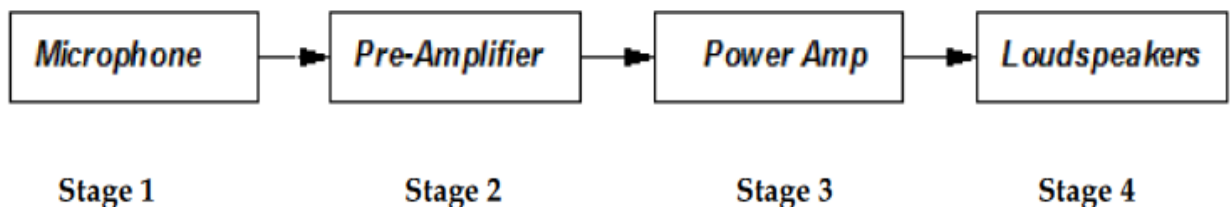


Fig 2.6: Block Diagram of a PA System

Stage 1: Microphone (Transducer)

The microphone converts sound waves into electrical signals that can be processed by the rest of the system. It is important that the microphone creates a faithful reproduction of the sound wave as an electrical signal – no distortion!

Stage 2: Pre-Amplifier

Its purpose is to take the small electrical signals from the microphone and increase the amplitude of the signal voltage.

Stage 3: Power Amplifier

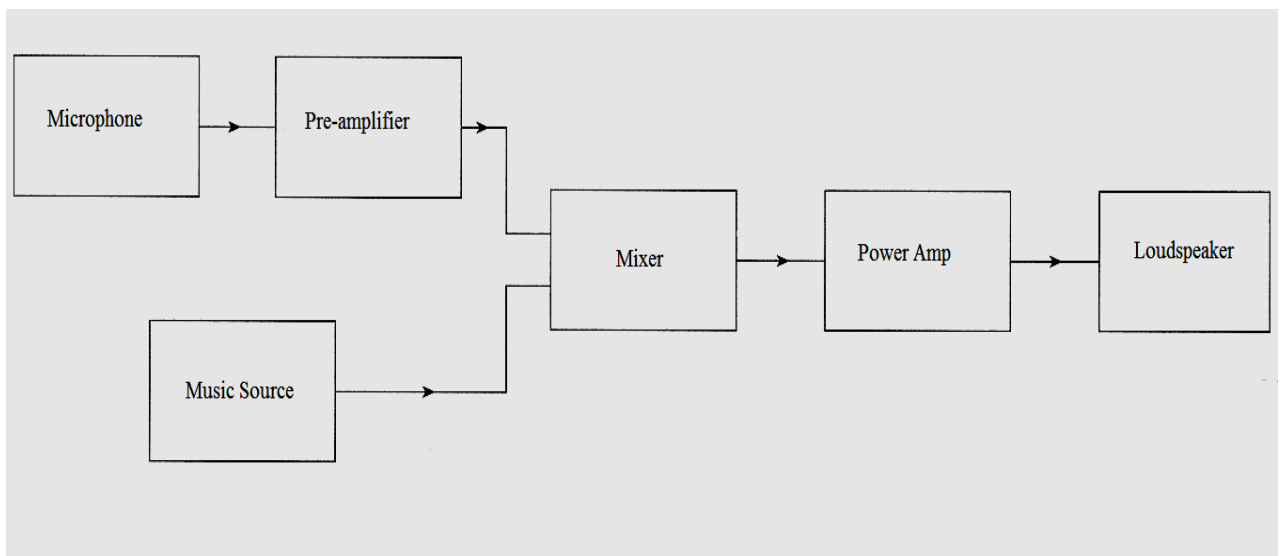
The power amplifier takes this enlarged voltage signal, and boosts the current so that it is strong enough to drive the loudspeaker.

Stage 4: Loud Speaker

The loudspeaker is the final part of the system where the electrical signal is transformed back into a sound wave.

If the system has carried out its function correctly, the emerging sound wave will be an undistorted but amplified version of the original.

A more sophisticated PA system would allow a number of inputs to be connected. For example, a band would have several microphone inputs and guitar pick-up inputs. These inputs would need to be faded in or out individually. Consider the following block diagram



It can be noticed that there are two additions to the simple PA system. The first is a music source and the second is a mixer.

Mixer: Its function is to add together electrical signals from microphones or pick-ups from electric guitars or backing tracks from a CD player.

Most music sources produce a much larger signal than a microphone and do not need a pre-amplifier. In a real system, each microphone would have its own pre-amplifier.

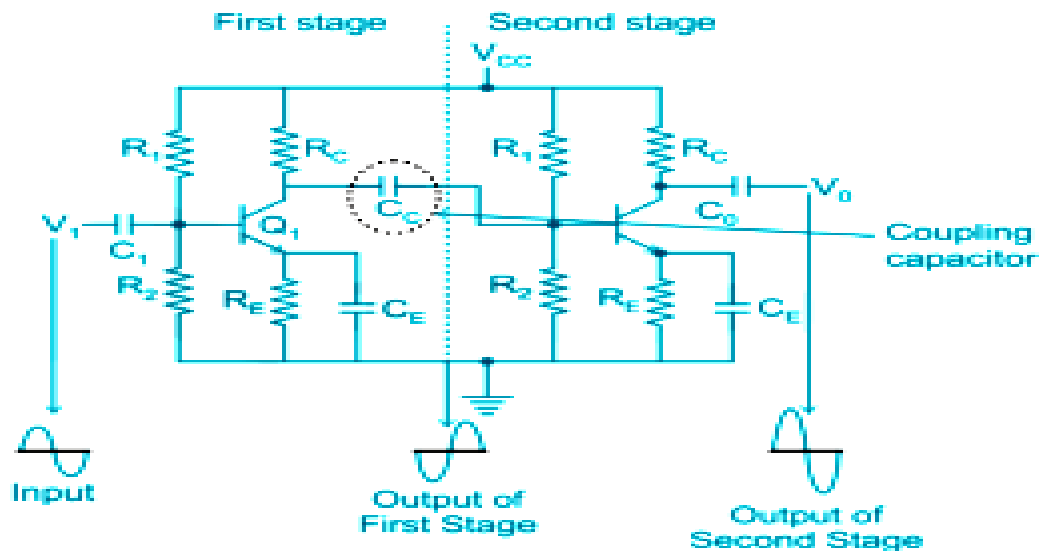
2.6 Common Emitter (RC coupled) amplifier

A transistor circuit containing more than one stage of amplification is defined as a multistage transistor amplifier. In this amplifier, a number of single amplifier stages are connected in a cascade arrangement. Cascading means the output of the first stage is connected to the input of the second stage through a suitable coupling device and so on. The coupling devices are generally capacitors, transformers, etc.

Construction of a Two stage RC Coupled Amplifier:

The below figure shows a two-stage R.C. coupled amplifier. Here the first stage is connected to the second stage through the coupling capacitor C_c

and resistor R_C . Therefore, this amplifier is called an RC-coupled amplifier. The resistors R_1 , R_2 form the self-bias and R_E form stabilization network. The bypass capacitor at the emitter offers a low reactance path to the signal. This method of coupling is widely used in cascade amplifier stages.



It can also drive low impedance loads because of its relatively low output impedance. Due to the non-linear properties of the transistors, this sort of amplifier can also be vulnerable to temperature variations and create distortion. To get the best performance, careful biasing, the use of suitable load resistors, and coupling capacitors are required.

Working of RC Coupled Amplifier:

When the input voltage is applied to the base of the first transistor through an input capacitor C_{in} , it appears as the amplified form across its collector load R_C . The output of the first stage is given to the base of the next stage through coupling capacitor C_c . The second stage does the further amplification of the signal.

The coupling capacitor C_c here helps in blocking the flow of DC components from the first stage to the second stage of the circuit. Thus, the biasing of the second stage will not get disturbed. The coupling capacitor has some reactance and hence there would be some loss of signal voltage due to the drop across this capacitor. This loss varies inversely with the frequency of the applied input signal. As capacitive reactance is inversely proportional to the frequency of the signal. $X_C = 1/2\pi fC$

We can notice that the total gain is less than the product of the gains from

individual stages. This is because when the second stage is made to follow the first stage, the effective load resistance of the first stage is reduced due to the shunting effect of the input resistance of the second stage. This reduces the gain of the stage which is loaded by the next stage.

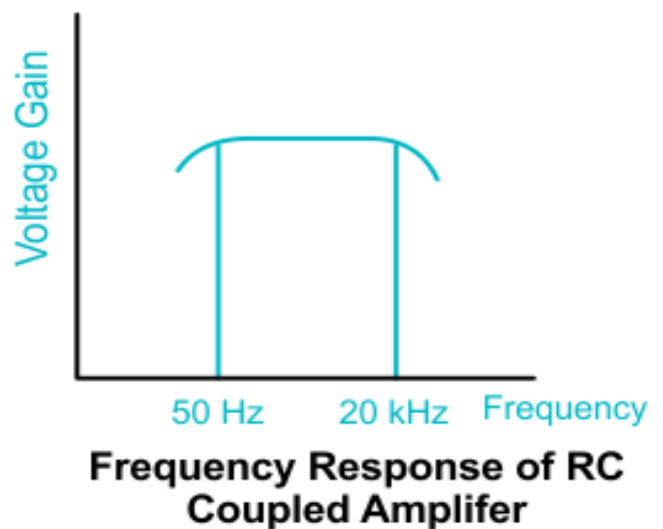
The ratio of output voltage V_O to the input voltage V_S is known as the amplification or voltage gain of the amplifier. Also called the loop gain, denoted as $A_v = V_O/V_S$

Frequency Response of RC Coupled Amplifier

Frequency response of RC coupled amplifier is uniform over the mid-frequency range of 50 Hz to 20 kHz. The voltage gain drops off at low < 50 Hz and high > 20 kHz frequencies. The frequency response curve of RC coupled with the amplifier is shown in Figure. The behavior of the amplifier is given below along with the bandwidth of an RC coupled amplifier between 50 Hz to 20 KHz.

Low Frequencies:

At low frequencies (< 50 Hz), the reactance of coupling capacitor C_C is quite high and hence very small part of the signal will pass from one stage to the next stage. The reactance of C_E is also high at low frequencies. This high reactance cannot shunt the emitter resistance R_E effectively. Due to these two factors, the voltage gain falls at low frequencies.



High Frequencies:

At high frequencies (> 20 kHz), the reactance C_C is very small and it behaves as a short circuit. This increases the loading effect of the next stage and serves to reduce the voltage gain. Also, the capacitive reactance of the base-emitter junction is low, and at high frequencies increases the base current. Due to these two reasons, the voltage gain falls at high frequencies.

Mid Frequencies:

At mid frequencies (50 Hz to 20 kHz), the voltage gain of the amplifier is constant. The effect of coupling capacitors in this frequency range is such as maintaining a uniform voltage gain. Thus, as the frequency increases in this range, the reactance of C_C decreases which tends to increase the gain. However, at the same time, lower reactance means higher loading of the first stage and hence lower gain. These two factors almost cancel each other, resulting in a uniform gain at mid-frequency.

2.7 Electronic Instrumentation:

Electronic Instrumentation is about the design, realization and use of electronic systems for the measurement of electrical and non-electrical quantities. Strongly related fields are measurement science and data acquisition. Each of these disciplines has a specific function in solving a measurement problem. The activity that is the basis of electronic instrumentation is “measuring”.

It is branch of engineering which deals with various types of instrument to record, monitor, indicate and control various physical parameters such as pressure, temperature, etc.

Block diagram of an electronic instrumentation system

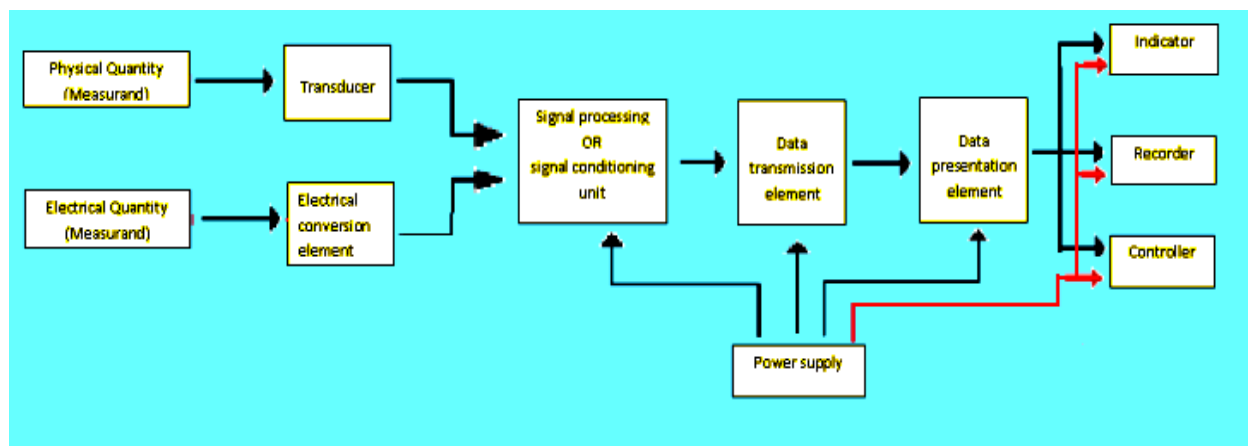


Fig 2.7: Block diagram of an electronic instrumentation system

Fig shows block diagram of generalized electronics instrumentation system, in which number of elements worked together to perform a desired function accurately i.e., to measure the measurand quantity and display it or record it.

Measurand (physical quantity or electrical quantity):

The physical or electrical quantity which is to be measurand is called as measurand. If the input to the instrumentation is parameter like pressure, force, level, strain, displacement, temperature, flow, velocity etc. then these parameters are physical measurand. These measurand are applied to the transducer element.

If the input is current, voltage and frequency then these parameters are called electrical measurand. These measurand are applied to the electrical conversion element.

Transducer and electrical conversion elements:

If the measurand is physical quantity then it is converted into equivalent electrical signal with help of element which is called transducer. Transducer is a device which convert one form of energy into another. (physical to electrical).

Most of the transducers are primary sensors which sense the measurand then

convert it into electrical signal with the help of conversion element.

If the measurand is already electrical signal like voltage, current or frequency then it is given to the electrical conversion element which convert the signal into more suitable form of signals.

Signal processing or signal conditioning:

The output from the transducer elements is given to the signal processing or signal conditioning elements. Signal conditioning is a process to modify the output of transducer so that it can be measurand, controlled and acceptable by next stages.

In electronic instrumentation system, filter, modulator, A/D converters, D/A converter, amplifiers, integrators, differentiators are the important signal conditioning circuits.

This stage is required to convert the transducer output into an electrical quantity suitable for proper operation of the last stage or indicators.

Data transmission element:

If the sensing element and data presentation element of the instrumentation system are away from each other (physically separated) in that case data transmission element is very important.

This element provides a transmission path for the modified signals to travel from transducer element to the rest of instrumentation elements like recorders, controllers, displays etc.

In electronic instrumentation system, typically the transmission path is a conducting line (i.e., electrical cables). In electronic instrumentation system some time radio link is used as a transmission path then the system is called telemetry system.

This element transmits the data to the remote located control room.

Data presentation element:

The signal from the data lines are provided to the data presentation element. This element converts the signal into such form that it can be presented by some visual or audible means.

Function performed by this stage may be demodulation, amplification, filtering, A/D conversion etc.

This element modifies the signals in such a way that the signals are accepted by recorders, displays, indicators, printers, announcing systems etc. The output of the data presentation element is provided to the recorders, controllers, and indicators as per the requirement of the user or operator or observer.

Output devices:

The last stage of instrumentation system is required to provide the information about the measurand for immediate reorganization by the operator whatever output is presented on indicator.

For example, if the system is simple instrumentation with display, then measured parameter is displayed directly on the display. If it is required to produce alarm for the over range then indicator or buzzer is used. If the recorder like strip-chart or X-Y recorder etc. To record the data.

If the system is control system in that case the measured data not only displayed or recorded but also compared with some reference value and control action is generated which is used to remove the error.

Power supply:

This is a common unit for all instrumentation system. This provides power to all elements working in the instrumentation setup.

If the transducer is active transducer (self-generating), in that case power supply is not required. For example, if that transducer is thermocouple or piezoelectric crystal.

For passive transducer power supply is required, for example if the transducer is thermistor or RTD, in that case power supply is important to the transducer block.

9. Assignments

S.No	Question	B L	C O
1	Describe the operation of a Bridge type Full-wave rectifier circuit along with circuit diagram. Draw the input and output wave forms.	3	1
2	Draw the circuit diagram of a RC coupled Common Emitter amplifier and explain its working principle. Also draw its frequency response.	2	1
3	Draw the block diagram of a Public Address system and explain its working principle.	3	1
4	Draw the block diagram of a Public Address system and explain its working principle.		

10. Part A- Question & Answers

S.No	Question & Answers	BL	CO
1	What is a Rectifier circuit?	1	2
2	Draw the circuit diagram of a Zener voltage regulator circuit.	1	2
3	List any two advantages of Bridge type full-wave rectifier circuit.	1	2
4	Draw the frequency Response of a Two Stage RC Coupled Amplifier	1	2
5	What is the purpose of filter in DC Power Supply system	1	2

11. Part B- Questions

S.No	Question	BL	CO
1	Draw the block diagram of a DC regulated power supply and explain each block with necessary waveforms.	2	2
2	Describe the operation of a Bridge type Full-wave rectifier circuit along with circuit diagram. Draw the input and output wave forms.	2	2
3	Draw the block diagram of a Public Address system and explain its working principle.	2	2
4	Draw the block diagram of an Electronic Instrumentation System and explain each block in detail.	2	2
5	Draw the circuit diagram of a RC coupled Common Emitter amplifier and explain its working principle. Also draw its frequency response.	2	2
6	Explain the working of Bridge rectifier with capacitor filter with necessary waveforms	2	2

12. Supportive Online Certification Courses

1. Basic Electronics -- Web course By Dr. Pramod Agarwal, Department of Electrical Engineering, IIT Roorkeer – 12 weeks
<https://nptel.ac.in/courses/117/107/117107095/#>
2. Basic Electronics -- Web course By Prof. Chitrlekha Mahanta, Department of Electronics and Communication Engineering, IIT Guwahati– 12 weeks.
<https://nptel.ac.in/courses/117/103/117103063/>

13. Contents Beyond the Syllabus**1. Conduction of Technical Quiz****2. Industrial visit to Amararaja Industries Pvt. Ltd.**

Students are exposed to various electronic components and electronic boards in real time measurements.

14. Prescribed Text Books & Reference Books**Text Book**

R.L. Boylestad and Louis Nashelsky, —Electronic Devices and Circuits,|| 9th Edition, Pearson, 2006.

